Chapter 3 Friday Flex

> Three tracks. No Friday activity is required for Week 4's new core instruction. Low-prep, quick reset. Same-object rule and measured-vs-inferred distinction are preserved throughout.

RECOVER

Field	Value
Time	flexible
Materials	unfinished Ramp graph, Drop It page, notebook inserts
Teacher prep	none
Student independence	independent/guided
Printable	as needed
Reusable	yes
Concepts reinforced	same object → faster → greater change; measured vs inferred

Options: - Finish the Ramp It Up! graph (Ramp Height vs. Average Cup Displacement). - Complete the Drop It evidence/explanation. - Notebook catch-up (Inserts 1–3). - Absent-student recovery (`student/recovery/03-ch3-absent-student-recovery.md`).

PRACTICE — The Fast & The Curious (tabletop data challenge)

Field	Value
Time	15–20 min
Materials	ramp data cards + clue strips (no physical locks)
Teacher prep	low — print/cut
Student independence	guided
Printable	yes
Reusable	yes

Field	Value
Concepts reinforced	data reading · same-object reasoning · measured vs inferred

Activity: Students work through 5 steps to "unlock" a 5-digit code (a word/numbers written on paper — no locks): 1. **Interpret** a small ramp table (1/2/3 books → cup displacement). 2. **Identify** the faster/slower trial using the SAME object. 3. **Match** evidence to the claim (faster → greater energy of motion). 4. **Correct** a false conclusion ("the heavier ball must be the reason"). 5. **Unlock** the code by combining one digit/letter from each step.

PRACTICE — Graph Error Detective

Field	Value
Time	10–15 min
Materials	sample graphs (with planted errors)
Teacher prep	low — print
Student independence	guided
Printable	yes
Reusable	yes
Concepts reinforced	correct axes · units · title · conclusions

Activity: Students find the errors in each sample graph: - wrong axis (labels swapped: displacement on x, height on y) - missing unit (cm) - bad title ("Speed vs. Energy") - unsupported conclusion ("we measured the energy")

PRACTICE — Same Object / Different Object Sort

Field	Value
Time	10 min
Materials	sort cards + 2 headers
Teacher prep	low — print/cut

Field	Value
Student independence	guided
Printable	yes
Reusable	yes
Concepts reinforced	same-object comparison (the only fair speed↔energy comparison)

Activity: Students sort scenarios into "SAME object (fair comparison)" vs. "DIFFERENT object (unfair comparison)".

PRACTICE — Claim + Evidence Match

Field	Value
Time	10 min
Materials	claim + evidence cards
Teacher prep	low
Student independence	guided
Printable	yes
Reusable	yes
Concepts reinforced	evidence supporting the speed↔energy relationship

Activity: Students match each evidence card to the claim: "Faster motion of the same object is associated with greater energy of motion."

EXTEND — Real-World Motion Analysis

Field	Value
Time	15 min

Field	Value
Materials	short verified clips (or written scenarios)
Teacher prep	verify clips in advance
Student independence	guided
Printable	no
Reusable	yes
Concepts reinforced	same-object speed↔energy in real contexts

Activity: For each example (e.g., a tennis ball serve vs. a gentle toss — SAME ball), identify: what changed · the faster/slower comparison · the evidence. External clips are candidates until verified.

Rule check

- [x] No new essential standard
- [x] Not required for Week 4's core instruction
- [x] Same-object rule preserved (no mass comparisons)
- [x] Measured-vs-inferred distinction preserved (no "we measured energy")
- [x] Low-prep, quick reset